

KMA Solaiman, Ph.D.

Assistant Teaching Professor, CSEE, UMBC
Director, H.A.R.M.O.N.I. Lab

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Academic Appointments

- 2023–Present **Assistant Teaching Professor**, Department of Computer Science and Electrical Engineering, University of Maryland Baltimore County (UMBC), MD, USA.
- 2016–2023 **Graduate Research & Teaching Assistant**, Department of Computer Science, Purdue University West Lafayette, IN, USA.
- 2014–2016 **Lecturer**, Department of Computer Science and Engineering, AUST, Dhaka, Bangladesh.

Education

- Summer 2023 **Ph.D. in Computer Science**, Purdue University, West Lafayette, IN.
Advisor: Bharat K. Bhargava (Thesis: *Multimodal Data Management in Open-world Environment*)
- Fall 2022 **M.Sc. in Computer Science**, Purdue University, West Lafayette, IN.
- July 2014 **B.Sc. in Computer Science and Engineering**, BUET, Dhaka, Bangladesh.

Research Areas and Applications

H.A.R.M.O.N.I. Lab

Human-Aligned, Resilient, Multimodal, Open-world, Novelty-informed Intelligence Lab

Research Areas:

- * Novelty-Aware and Adaptive Artificial Intelligence,
- * Multimodal Information Retrieval and Data Fusion, and
- * Deployment-Aware AI under Constrained Sensing

Applications: Cyber-Physical and Smart-grid Systems, Emergency Triage and Healthcare Decision Support, Public Safety and Operational AI Systems.

Current Projects: NovaSGrid, FemmlR, BiasLab, TRIAGE-M, BoardVision.

Collaborators: UMBC Center for AI, UMMC, UMB School of Pharmacy, Purdue University.

Lab Website: <https://ksolaiman.github.io/harmoni-lab/>

Prior Research Ecosystem. During doctoral and early-career research, my work involved collaborations across academia, government, and industry, including MIT, Purdue University, University of Michigan, USC–ISI, Northrop Grumman, and IDA, shaping my approach to large-scale, deployment-driven AI research.

Teaching Experience

- 2023 – Present **University of Maryland, Baltimore County**, Assistant Teaching Professor.
- CMSC 678: Graduate Machine Learning (≈35 students) Fall'24, '25
 - CMSC 478: Machine Learning (≈70 students) Fall'23, '24, '25, Spring'24, '25, '26
 - CMSC 471: Artificial Intelligence (≈80 students) Fall'23, '24, '25, Spring'24, '25, '26
- 2016 – 2023 **Purdue University**, Teaching Assistant.
- Courses: CS 180, CS 251, CS 448, CS 543, CS 536 (undergraduate and graduate levels)
 - Subjects: Programming, Data Structures, Database Systems, Simulation, Computer Networks
- Fall 2019 **Purdue University**, Guest Lecturer.
- CS 590: Situational Awareness, Adversarial ML, and Explainable AI Graduate Course
 - CS 547: Information Retrieval Graduate Course
- 2014 – 2016 **Ahsanullah University of Science & Technology, United International University**.
- Primary Instructor for multiple courses from 100 to 400 level, with classes of ≈100 students.

Course Evaluation Highlights:

- All taught courses have demonstrated sustained performance in the 4.5–4.8 range across multiple semesters, based on consistently strong participation in course evaluations.

Student Mentoring and Advising

- * Mentoring has led to peer-reviewed publications, a departmental best paper award, and successful student placements in industry and graduate programs.
- * Academic advising emphasizes career guidance and sustained support through challenging periods.

M.Sc. Thesis Advisor and Committee Chair

- Sumya Hamid Oishe (UMBC). *Comparative Analysis of Anomaly Detection Models for Irregular and Incomplete Time Series in Cyber-Physical Systems*.

Ph.D. Committee Member

- Saad Rahman (UMBC). *Resilient Computation in Resource-constrained Reconfigurable Platforms*.
- Rachit Saini (UMBC). *Embedded Automotive IDS for In-vehicle Communication Networks*.

M.Sc. Committee Member

- Rachit Saini (UMBC). *CAN IDS on FPGA Platform*.
- Venkata Sundarapu (UMBC). *PLL and DLL Design Using CMOS Phase Frequency Detector*.

Current H.A.R.M.O.N.I. Lab Students/Researchers and Projects

- Joshua Sebastian, Karma Tobden – TRIAGE-M
- Salma Negmeldin – TRIAGE-M/VKIL
- David Majekodunmi – NovaSGrid/Continual Resiliency

Past H.A.R.M.O.N.I. Lab Students/Researchers and Projects

- Adam Sayyed – MStolert
- Sumya Hamid Oishe – NovaSGrid
- Brandon Hill – BoardVision

Past Mentored Students

- UMBC, MIT, METU, and Purdue graduate and undergraduate researchers
(Selected Outcomes: Meta, Cisco, DoD, Five9, WePay; Cambridge, JHU, TUM)

Selected Publications

* Full list available on [Google Scholar](#)

** J = Peer-reviewed Journal Article, C = Peer-reviewed Conference paper, W: Peer-reviewed Workshop/Symposium Paper, P = Preprint.

- IEEE ICHI 2026 **KMA Solaiman**, Joshua Sebastian, and Karma Tobden,
[C3] *Benchmarking Early Deterioration Prediction in Emergency Triage Under Constrained Sensing*,
Accepted, IEEE ICHI 2026.
<https://arxiv.org/abs/2602.20168>
- EdgeAI 2026 **KMA Solaiman**, S. Islam, R. Oliveira, B. Bhargava,
[W4] *Pre-Deployment Complexity Estimation for Federated Perception Systems*,
Accepted, Edge AI Research Symposium 2026, San Diego, CA..
<https://ksolaiman.github.io/files/publications/FedComplex.pdf>
- ICML 2025 **KMA Solaiman**,
[W3] *BiasLab: Explainable Political Bias Detection with Dual-Axis Annotations and Rationale Indicators*,
Presented at the Workshop on Models of Human Feedback for AI Alignment (MoFA), ICML 2025.
<https://arxiv.org/pdf/2505.16081>

- arXiv 2026 Brandon Hill and **KMA Solaiman**,
[P1] *BoardVision: Deployment-ready and Robust Motherboard Defect Detection with Ensemble*, Submitted at WACV Applications 2026, Presented at UMBC CSEE Research Day 2025.
Preprint: <https://arxiv.org/abs/2510.14389v1>
- Computer 2022 **KMA Solaiman**, Tao Sun, Alina Nesen, Bharat Bhargava, and Michael Stonebraker,
[J1] *Applying Machine Learning and Data Fusion to the Missing Person Problem*,
IEEE Computer, Volume: 55, Issue: 6, June 2022.
- AAAI 2022 **KMA Solaiman** and Bharat Bhargava,
[W2] *Open-Learning Framework for Multi-modal Info. Retrieval with Weakly Supervised Joint Embedding*,
AAAI Spring Symposium on Designing Artificial Intelligence for Open Worlds, March 2022.
- AAAI 2022 **KMA Solaiman** and Bharat Bhargava,
[W1] *Measurement of Novelty Difficulty in Monopoly*,
AAAI Spring Symposium on Designing Artificial Intelligence for Open Worlds, March 2022.
- IEEE TransAI 2021 Alina Nesen, **KMA Solaiman** and Bharat Bhargava,
[C2] *Dataset Augmentation with Generated Novelty*,
IEEE International Conference on Transdisciplinary AI (TransAI), 2021.
- IEEE ICIEV 2013 **KMA Solaiman**, MM Rahman, and Nashid Shahriar,
[C1] *AVRA BANGLADESH: Collection, Analysis & Visualization of Road Accident Data in Bangladesh*,
IEEE International Conference on Informatics, Electronics & Vision (ICIEV), 2013.

Grants, Proposals & Contributions

- 2026-2027 **Maryland BiLD Grant, \$75K**, PI, LOI Submitted
Fusion of Opportunistic Vital Monitoring and Structured Triage for Decision Support.
- 2024-2027 **Department of Energy (DoE), \$1.35M/year**, Key Personnel, Submitted
A Reformable Cyber-Physical System with Assured Continuous Operation Under Cyber-Attacks.
- 2026-2028 **Hrabowski Fund for Innovation SoTL, \$10K**, PI, Submitted
Teaching Fairness through Annotation: Evaluating Bias Understanding in Applied AI Courses.
- 2025-2026 **Maryland Open Source Textbook (M.O.S.T.) Institutional OER Grant, \$20K**, PI, Submitted
Launching a Modular, Open Curriculum in Data & AI Literacy for Non-Majors.
- 2025 **NAIRR AI Education Fellowship**, Applied
National fellowship supporting scalable AI education and infrastructure.
- 2024-2025 **University of Maryland Baltimore (UMB) ICTR, \$50K**, Co-Investigator, Submitted
Optimizing Pharmacist Team-Integration for ICU Patient Management (OPTIM).

Active Contributions on Existing Grants as Graduate Researcher

- 2019-2024 **Sandia National Lab STARCS, Northrop-Grumman REALM Consortium, DARPA SAIL-ON**
Worked on resilient and multimodal learning models, applications in Resilient Space and deployment Platforms.

Service & Leadership

Institutional Service

- **Graduate Admissions Committee**, UMBC (2025-2026)
Evaluated and selected graduate applicants for the CSEE program.
- **Faculty Hiring Committee**, UMBC (2024-2025)
Participated in evaluation and selection of faculty candidates.
- **Undergraduate Admissions Committee**, UMBC (2023-2024)
Reviewed course equivalencies, supported ABET-related evaluation processes, and contributed to undergraduate curriculum alignment.
- **Curriculum Development, NSA & DPS**, UMBC (2024)
Developed professional-facing data science and AI curriculum materials.

- **Undergraduate Thesis Committee**, AUST, Bangladesh (2016)
- **Undergraduate Curriculum Accreditation**, AUST, Bangladesh (2016)

Program Committee Member/ Reviewer

- ICML MoFA (2025); ICDE (2027, 2025); XAI-2026; ECML-PKDD (2021–2022); IEEE PIMRC (2019)

Instructional Innovation & Curriculum Development

- Developed SoTL-aligned instructional proposal for teaching fairness and interpretability in AI.
- Designed modular AI content for non-specialist learners.
- Contributed to professional-facing curriculum development (NSA/DPS).
- Proposed open-access AI literacy curriculum (OER initiative).
- Initiated a multi-institutional AI education proposal (NAIRR AI Education Fellowship), aimed at scaling modular, human-aligned AI curricula and datasets through shared infrastructure and cross-institution collaboration through USM.

Selected Talks & Outreach

- Feb 2025 **Modern Database Systems**
East Tennessee State University, Johnson City, TN.
Invited Talk, Host: Chelsie Dubay
- Nov 2023 **Shaping the Future of Computer Science**
Purdue University, West Lafayette, IN.
Invited Talk, Host: H.E. Dunsmore
- May 2022 **Detect, Characterize, and Accommodate Novelties in AI systems**
International Semantics Intelligence Conference (ISIC).
Invited Talk, Co-presented with Bharat Bhargava
- July 2021 **Adaptable AI Systems to deal with Novelties and Attacks**
Artificial Intelligence Campaign Tech Talk.
Invited Talk, Co-presented with Bharat Bhargava, Host: Reid Hyland
- May 2021 **Information-Theoretic Approach to Measurement of Novelty Difficulty in Monopoly**
DARPA SAIL-ON NWG (Novelty Working Group).
Invited Talk, Hosts: Joshua Alspector and Pat Langley
- Jan 2020 **Situation Knowledge on Demand (SKOD)**
Jet Propulsion Laboratory (NASA), Pasadena, CA.
CDER Invited Talk, Hosts: Arun Viswanathan and Jeremy Pecharich

Awards and Honors

- 2026 **Best Paper Award**, CSEE Research Day 2026, UMBC (TriageM; undergraduate-led project).
- 2026 **Undergraduate Presentation Grants**, Undergraduate Research & Prestigious Scholarships, UMBC. Joshua Sebastian in **IEEE ICHI 2026**. \$750.
- 2025 Active Learning, Inquiry Teaching (ALiT) Certificate, UMBC Faculty Development Center. \$500.
- 2023 **Supplement for Undergraduate Research Experiences (SURE)**, Dept. of CSEE, UMBC. \$1500.
- 2018 Graduate School Summer Research Grant, Purdue University. *Summer Research Support*.
- 2009-2012 University Merit Scholarship, Bangladesh University of Engineering and Technology.
- 2008-2011 University Stipend and Dean's List, Bangladesh University of Engineering and Technology.

Open Source Software and Demo

- 2025 **BiasLab: Political Bias Annotation Corpus (ICML MoFA 2025).**
Dataset (<https://zenodo.org/records/15571668>)

- 2022 **Find-Them: Video demonstration of the system prototype (IEEE Computer 2022).**
https://youtu.be/hJ_jtLQUIXo
- 2020 **Surveillance Video Querying engine (SurvQ) : (SIGMOD 2020).**
<https://github.com/skod-ng/>
- 2019 **Situational Knowledge on Demand (SKOD) : Open-source framework for multimodal ingestion and knowledge graph construction (VLDB 2019).**
<https://github.com/purdue-gask>

Teaching Areas & Interests

Artificial Intelligence, Machine Learning, Data Mining, Database Systems, Information Retrieval, Responsible and Deployment-Aware AI, AI for Cyber-Physical Systems